

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 2nd Term Examination, 2020
Department of Materials Science and Engineering

Ch 1227
Organic Chemistry

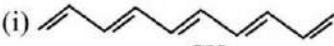
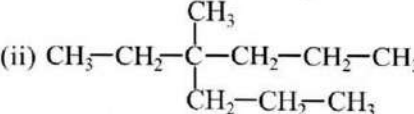
Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) State the resonance theory to explain the structure of a molecule or ion. If a molecule shows resonance structures, how can you decide which resonance structure is more stable than another? Explain with a suitable example. (12)
- (b) Explain the effect of hybridization of the central atom on the bond lengths of a molecule. (10)
- (c) C=C double bond rotation is not possible. Why? (08)
- Q 2. (a) Which of the following carbocations would you expect more stable? Why? (10)
- (i) R_3C^+ , (ii) R_2HC^+ , (iii) RH_2C^+ , and (iv) H_3C^+
- (b) Which S_N1 reaction of each pair would you expect to take place more rapidly? Why? (12)
- (i) $R_3CCl + H_2O \longrightarrow R_3C(OH) + HCl$
- (ii) $R_3CI + H_2O \longrightarrow R_3C(OH) + HI$
- (c) Write IUPAC name of the following compound: (08)
- (i) 
- (ii) 
- Q 3. (a) How can you prove that the following compounds are aromatic? (12)
- (i) Benzene and (ii) Phenol
- (b) What is aromaticity? What are the necessary conditions to be aromatic? (10)
- (c) CH_3Cl is more polar than CH_3OH . Is it correct or not? Explain. (08)

Section B

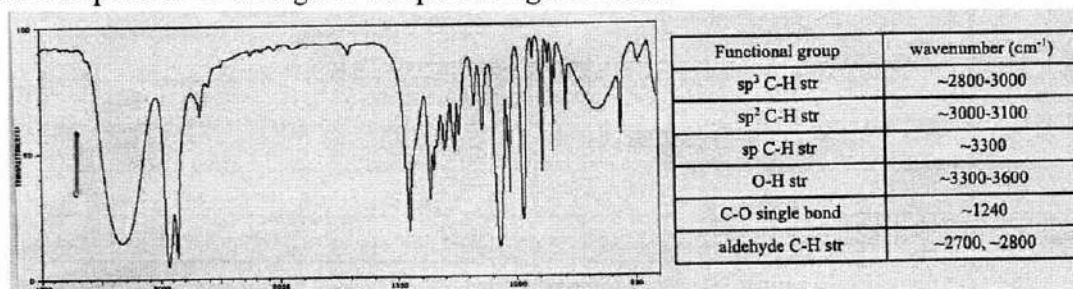
(Answer ANY TWO questions from this section in **answer script B**)

Q 4. (a) Which of the following molecules do not absorb in the Infra-red region and why? (08)

(i) O₂ (ii) CO₂

(b) Calculate the stretching vibration for C=O bond. [Here, Mass of the C atom = 20×10^{-24} g, (10)
Mass of O atom = 26.6×10^{-24} g, $K = 5 \times 10^5$ g sec⁻², $C = 3 \times 10^{10}$ cm/sec]

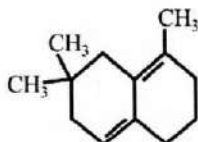
(c) The IR spectrum of an organic compound is given below (12)



(i) What type of compound represent the above spectrum?

(ii) Is it saturated or unsaturated?

Q 5. (a) Calculate the absorption maximum of the following compounds according to Woodward-Fieser rules: (08)



(b) Write down the chemical reactions and uses of cellulose nitrates derivative. (12)

(c) What is meant by isoelectric point? Draw the chemical structures of glycine at pH = 2 and pH = 12. (10)

Q 6. (a) What are conducting polymers? Explain the mechanism of conduction in polyacetylene. (12)

(b) Explain the free radical mechanism of addition polymerization by taking ethylene as an example. (12)

(c) Write down three differences between thermoplastic and thermosetting polymer. (06)

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 2nd Term Examination, 2020
Department of Materials Science and Engineering

Math 1227
Differential Equations and Geometry

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) Define order and degree of differential equations with examples. (16)
Solve, $(1 + y^2)dx = (\tan^{-1}y - x)dy$
(b) Find the general solution of $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$ (14)
- Q 2. (a) Determine the general solution of $\frac{d^2y}{dx^2} - y = \cos x + (x^2 + 1)e^x$ (16)
(b) Solve $x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} + 2y = x \log x$ (14)
- Q 3. (a) Determine the Laplace transform of $F(t)$, where (08)
$$F(t) = \begin{cases} t & \text{for } 0 < t < 2 \\ 3 & \text{for } t > 2 \end{cases}$$

(b) Find the inverse Laplace transform of $\frac{(s + \frac{1}{2})e^{-\pi s}}{s^2 + s + 1}$ (09)
(c) Solve the differential equation $Y'' + Y = t$, $Y(0) = 1$, $Y'(0) = -2$ by using Laplace transform. (13)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) Determine the cylindrical polar coordinates and spherical polar coordinates, whose cartesian coordinates is $(\sqrt{3}, 1, 2\sqrt{3})$. (12)
(b) Define direction cosines of a line. (18)
If the direction cosines of two lines are connected by the relations $2l + 2m - n = 0$ and $lm + mn + nl = 0$, then find the direction cosines of two lines.

- Q 5. (a) Find the equation of the plane through the points $(1, 2, 1)$ and $(2, 3, -1)$ and perpendicular to the plane $x + 2y + 2z - 5 = 0$. (10)
- (b) Find the distance between the plane $x - 2y - 3z + 10 = 0$ and the point $(1, 2, -1)$. (07)
- (c) Find the angle between the line $2x + 4y - 2z + 3 = 0 = 4x - 2y + 6z + 5$ and the plane $5x - 4y + 3z - 5 = 0$ (13)
- Q 6. (a) Find the shortest distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$; $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ state whether the lines are coplanar or not. (15)
- (b) Find the two tangent planes to the sphere $x^2 + y^2 + z^2 - 4x + 2y - 6z + 5 = 0$ which are parallel to the plane $2x + 3y - z = 0$. (15)

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 2nd Term Examination, 2020
Department of Materials Science and Engineering

ME 1227
(Engineering Mechanics)

Time: 1.5 hours

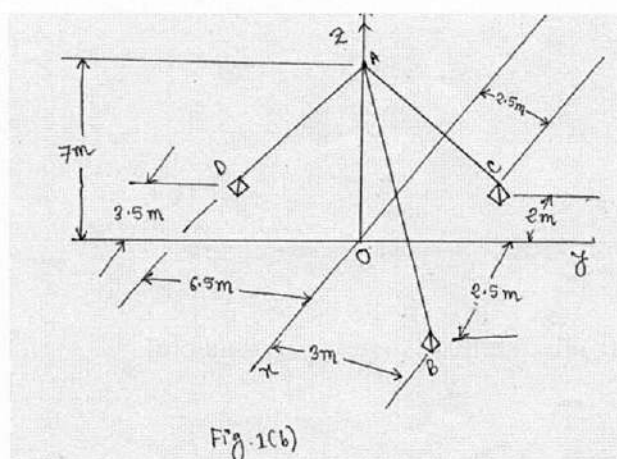
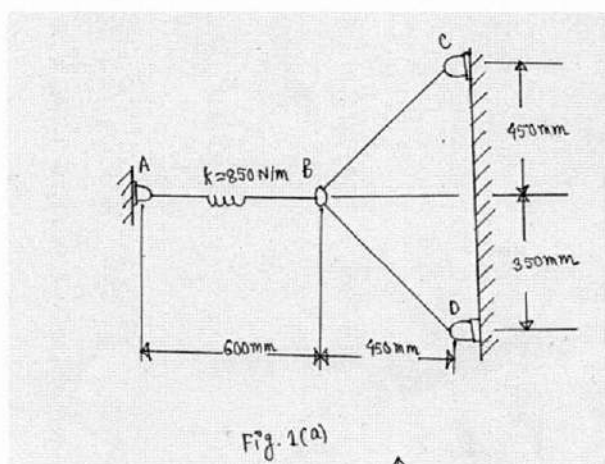
Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

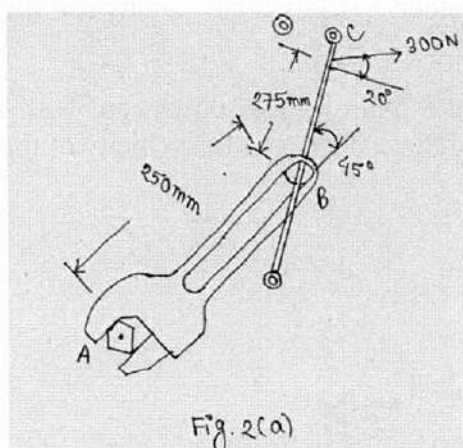
Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

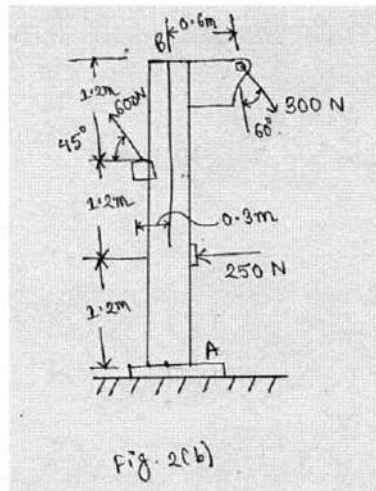
- Q 1. (a) The spring has a stiffness of $k=850\text{N/m}$ and an unstretched length of 300mm. Determine the force in cables BC and BD when the spring is held in the position shown in Fig. 1(a). (14)
- (b) If cable AB is subjected to a tension of 850N, determine the tension in cables AC and AD. (16)
Shown in Fig. 1(b).



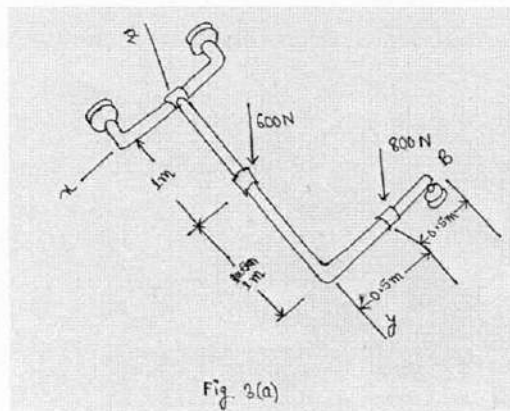
- Q 2. (a) The connected bar BC is used to increase the lever arm of the crescent wrench as shown. (13)
If the applied force is 300N, determine the moment produced by this force about the bolt at A. Shown in Fig. 2(a).



- (b) Replace the force system acting on the post by a resultant force and specify where its line of action intersects the post AB, measured from point B. Shown in Fig. 2(b). (17)



- Q 3. (a) Determine the support reactions at the smooth collar A and the normal reaction at the roller support B. Shown in Fig. 3(a). (20)

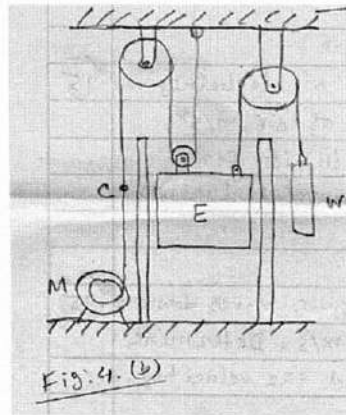


- (b) Explain D'Alembert's principle. (10)

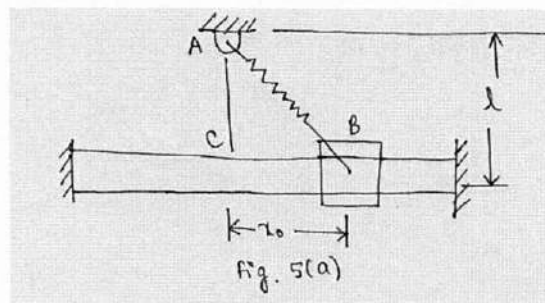
Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

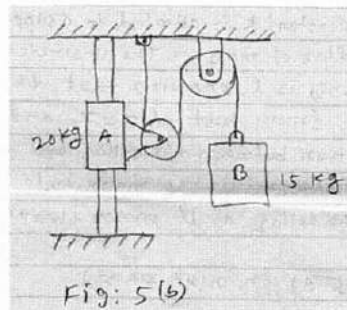
- Q 4. (a) A truck travels 220m in 10s while being decelerated at a constant rate of 0.6m/s^2 . (15)
Determine (i) its initial velocity, (ii) its final velocity and (iii) the distance traveled during the first 1.5s.
- (b) The elevator shown in the figure moves downward with a constant velocity of 5m/s. (15)
Determine the velocity of the cable C and the velocity of the counterweight W. Shown in Fig. 4(b)



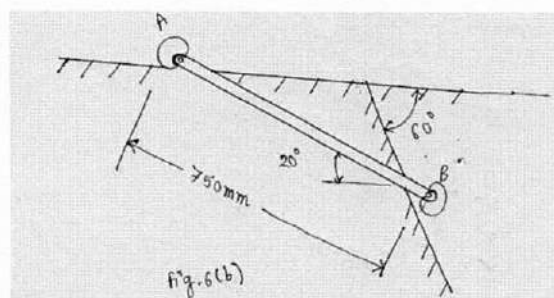
- Q 5. (a) A spring AB of constant k is attached to a support at A and to a collar of mass m . The unstretched length of the spring is l . Knowing that the collar is released from rest at $x=x_0$ and neglecting friction between the collar and the horizontal rod, determine the magnitude of the velocity of the collar as it passes through point C. Shown in Fig. 5(a) (15)



- (b) The system shown is released from rest. Determine the time it takes for the velocity of A to reach 1m/s . Neglect friction and the mass of the pulleys. Shown in Fig. 5(b). (15)



- Q 6. (a) Why governor is used in stationary engine? Discuss the classification of Cam. (10)
- (b) Small wheels have been attached to the ends of rod AB and roll freely along the surfaces shown. Knowing that wheel A moves to the left with a constant velocity of 1.5m/s , determine (i) the angular velocity of the rod and (ii) the velocity of end B of the rod. Shown in Fig. 6(b). (20)



Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 2nd Term Examination, 2020
Department of Materials Science and Engineering

Ph 1227
(Magnetism and Nuclear Physics)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) Explain the justification of negative value of packing fraction. (08)
(b) Write down proton-proton cycle and Carbon-Nitrogen cycle for fusion reaction. (12)
(c) Calculate the energy (in kWatt) released from 1 gm of ²³⁵U. Assume 200 MeV energy is released per fission reaction. (10)
- Q 2. (a) Discuss the biological effect of radiation. (10)
(b) Write down the application of radio-activity. (10)
(c) Some amount of radio-active substance of half-life 30 days is spread inside a room. Consequently, the level of radiation inside the room became 50 time the permissible level for normal occupancy of the room. After how many days the room would be safe for occupation? (10)
- Q 3. (a) Explain the terms: (i) yield point (ii) elastic limit (iii) breaking stress (09)
(b) Prove that strain energy per unit volume = $\frac{1}{2} \times \text{Bulk modulus} \times (\text{Volume strain})^2$ (12)
(c) Find the energy stored up in a wire 12 meters long and 1.2 mm in diameter which has been stretched by 3/10 mm, due to a load of 12 kg ($g=300\pi$) (09)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) Explain the term piezoelectricity with necessary examples. (05)
- (b) Explain space quantization and spin quantization in vector model with necessary diagram. (15)
Hence write down all quantum number associated with this model with their mathematical notations.
- (c) Calculate the Lande's g-factor of the hypotheticalal $3p_2$ and $1S_0$. (10)
- Q 5. (a) Explain the terms: (i) Neel temperature (ii) Curie law and (iii) Heushen alloy. (09)
- (b) Find the magnetic field due to a long straight wire carrying a current using Biot-Savart law. (12)
- (c) The of crystal sodium chloride has the static dielectric constant of 2.6 and optical index of refraction 1.5. Calculate the percentage contribution of ionic polarizability. (09)
- Q 6. (a) Explain the concept of dielectric and dielectric constant. Derive expression for dielectric constant with and without taking into account local fields. (15)
- (b) What is origin of magnetism? How are the different types of magnetic material classified? (08)
- (c) State Maxwell's four equations of electro-magnetism in free space. (07)

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 2nd Term (Regular) Examination, 2020
Department of Materials Science and Engineering

HUM 1227
(Technical English)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) Make sentence with the following structures using the words given in brackets. (10)
- i) Subj. + Intransitive Verb + Adv. of manner. (Walk as Verb)
 - ii) Subj. + Transitive Verb + Direct Object. (Drink as Verb)
 - iii) Neither + Subj. + nor + Subj. + Verb + Adv. (Play as Verb)
 - iv) Subj. + Verb + as + Subj. + Verb + Obj. (Complete and Suggest as Verb)
 - v) That + Subj. + Verb + Adv. + Verb + Adj. Complement. (think and is as Verb)
- (b) Transform the following words as asked in brackets and make sentence with the transformed words. (10)
- Alleviate (into noun), Intimacy (into adj.), Rich (into verb), Long (into noun), Assumption (into verb)
- (c) Expressing the following notions/emotions, make sentences. (10)
- i) Annoyance ii) Condolence iii) Good Wish
 - iv) Seasonal Greeting v) Introduction
- Q 2. (a) Make WH questions with the underlined words of the following sentences. (10)
- i) Soma reads with a great interest.
 - ii) Liza is five feet and two inches tall.
 - iii) Amir watched the scene.
 - iv) The bus is running at a speed of 30 Kilometers.
 - v) The book is interesting to read.
- (b) Make sentences using the following words as directed (10)
- Close (as noun); Close (as verb); Down (as adjective); Down (as adverb); Down (as preposition)
- (c) Make use of the following modals in sentence as asked in brackets. (10)
- i) May (To allow someone for something)
 - ii) Should + have + past participle of verb (To express past action, which was not implemented)
 - iii) Dare (To express indulgence)
 - iv) Would (To express past irregular habit)
 - v) Be + verb + ing (To express an arrangement)

Q 3. (a) Transform the following sentences asked in brackets. (10)

- i) Soma, a talented student, is hard working. (Complex)
- ii) Nasim working hard progressing life. (Complex)
- iii) As Liza is absent, let's avoid the issue. (Simple)
- iv) He works hard so that he can enrich in life. (Simple)
- v) Inspite of living honest nobody likes him. (Compound)

(b) Complete the following sentences with clauses as asked in brackets. (10)

- i) _____ is right (Noun Clause)
- ii) _____ he can't walk quickly. (Adverbial Clause)
- iii) We named the house _____. (Noun Clause)
- iv) Pip, _____, is a doctor. (Adj. Clause)
- v) He works hard _____. (Adv. Clause)

(c) Correct the following sentences. (10)

- i) Chairman is supposed to chair in the meeting.
- ii) He that well works praiseworthy is.
- iii) He always tells the truth.
- iv) He is sick of fever.
- v) When he is ill, he will not come in the meeting.

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

Q 4. (a) Write a paragraph on "Good Citizenship". (18)

(b) Amplify the idea – 'Youth is full of pleasure, age is full of care.' (12)

Q 5. Suppose you are a job seeker, prepare a CV with an application. (30)

Q 6. Write a free composition on any **one** of the following. (30)

- a) Global Warming : Bangladesh is a victim
- b) Corruption in Bangladesh